

Neglect rehabilitation by prism adaptation: Different procedures have different impacts

Elisabetta Làdavas^{a,b}, Silvia Bonifazi^c, Lorena Catena^d, Andrea Serino^{a,b,*}

^a Dipartimento di Psicologia, Università degli Studi di Bologna, Italy

^b Centro studi e ricerche in Neuroscienze Cognitive, Polo Scientifico-Didattico di Cesena, Italy

^c Clinica di Neuroriabilitazione, Azienda Ospedali Riuniti Lancisi-Salesi-Umberto I, Ancona, Italy

^d Istituto di Riabilitazione Santo Stefano, Porto Potenza Picena, Macerata, Italy

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ABSTRACT

Several studies point to prism adaptation as an effective tool for the rehabilitation of hemispatial neglect. However, some recent reports failed to show a significant amelioration of neglect after prism adaptation as compared to control treatments. This apparent contradiction might reflect important differences in the procedures used for treatment. Here we compare the effects of two treatments (performed for 10 sessions, over 2 weeks) in two groups of patients, based either on a **Terminal (TPA) or a Concurrent (CPA) prism adaptation procedure**. During TPA only the final part of the pointing movement is visible and prism adaptation relies most strongly on a strategic recalibration of visuomotor eye–hand coordinates. In contrast, during CPA the second half of the pointing movement is visible, and thus adaptation mainly consists of a realignment of proprioceptive coordinates.

The present results show that both TPA and CPA treatments induced a greater improvement of neglect as compared to a control treatment of pointing without prisms. However, neglect amelioration was **higher for patients treated with TPA than for those treated with CPA**. At the same time, the **TPA treatment induced a stronger deviation of eye movements toward the left, neglected, field as compared to the CPA treatment**. Interestingly, in TPA patients the visuomotor and oculomotor effects of the treatment were directly related to the patients' ability to compensate for the optical deviation induced by prism during pointing (i.e., Error reduction effect).

In summary, prism adaptation seems particularly effective for the recovery of visuo-spatial neglect when conducted with a procedure stressing a correction of visuomotor eye–hand coordinates, i.e., with a TPA procedure. The present observations may help to better understand the mechanisms underlying prism-induced recovery from neglect and the procedural basis for some of the contradictory results obtained when using this rehabilitative strategy.

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1. Introduction

Unilateral spatial neglect ('neglect') is a failure to report, respond, or orient to stimuli that are presented contralateral to a brain lesion (Heilman & Valenstein, 1979). Neglect symptoms range from a slowing of responses to contralesional stimuli to a complete lack of awareness of the contralesional half of space, at which point, patients behave as if that half of the world does not exist. A left neglect syndrome is frequently observed in right brain-damaged patients and is often severe enough to constitute a major handicap (Buxbaum et al., 2006; Denes, Semenza, Stoppa, & Lis, 1982; Milner & McIntosh, 2005). Thus, a rehabilitation protocol that would ame-

liorate neglect symptoms would also have a great impact on clinical outcome. In this respect, prism adaptation seems a particularly promising rehabilitative procedure (see Chokron, Dupierri, Tabert, & Bartolomeo, 2007; Luaute, Halligan, Rode, Rossetti, & Boisson, 2006a; Luaute, Halligan, Rode, Jacquin-Courtois, & Boisson, 2006b; Pisella, Rode, Farne, Tilikete, & Rossetti, 2006; Rode, Klos, Courtois-Jacquin, Rossetti, & Pisella, 2006; Striener & Danckert, 2010, for reviews), since this technique has been shown to produce some improvement in a wide range of neglect symptoms in the visual (e.g. Farne, Rossetti, Toniolo, & Ladavas, 2002; Frassinetti, Angeli, Meneghello, Avanzi, & Ladavas, 2002; Rode, Rossetti, & Boisson, 2001; Rode, Pisella, Rossetti, Farne, & Boisson, 2003; Rossetti et al., 1998; Serino, Angeli, Frassinetti, & Ladavas, 2006; Serino, Bonifazi, Pierfederici, & Ladavas, 2007; Serino, Barbani, Rinaldesi, & Làdavas, 2009), somatosensory (Dijkerman, Webeling, ter Wal, Groet, & van Zandvoort, 2004; Maravita et al., 2003; Tilikete et al., 2001) and auditory (Jacquin-Courtois et al., 2010) domains. Repeated sessions

* Corresponding author at: Centro studi e ricerche in Neuroscienze Cognitive, Via Brusi 20, 47123 Cesena, Italy. Tel.: +39 0547338951; fax: +39 0547338952.

E-mail address: andrea.serino@unibo.it (A. Serino).

of prism adaptation over two weeks induce a long-lasting improvement not only in standard clinical tests of neglect but also in more ecological measures (Frassinetti et al., 2002; Jacquin-Courtois, Rode, Pisella, Boisson, & Rossetti, 2008; Shiraishi, Yamakawa, Itou, Muraki, & Asada, 2008).

The prism adaptation procedure consists of the repetition of a series of pointing movements while wearing prismatic lenses that shift the visual field toward the right. To compensate for this optical displacement, the patient has to orient his/her pointing movement toward the left of visual space, resulting in a leftward drift of sensorimotor coordinates. The beneficial effects of prism adaptation on several neglect symptoms shows that visuomotor adaptation can modify high-level spatial representations, suggesting a direct link between plasticity of the sensorimotor system and space representation.

Despite the positive results obtained in many studies, some authors did not find a significant amelioration of neglect symptoms after prism adaptation (e.g. Nys, de Haan, Kunneman, de Kort, & Dijkerman, 2008; Rousseaux, Bernati, Saj, & Kozłowski, 2006; Turton, O'Leary, Gabb, Woodward, & Gilchrist, 2009). However, these contradictory results may be due to procedural differences in the prism adaptation procedures used by different laboratories. These procedural differences, and their possible impact on the recovery of neglect, are examined here within the context of their ability to induce plasticity in sensorimotor and spatial representations.

Recent findings from our laboratory have shown that visuomotor training consisting of pointing toward visual stimuli even without wearing prismatic goggles (i.e., neutral pointing) induced a moderate improvement of neglect symptoms. However, the improvement was significantly stronger and became clinically relevant (i.e., above the cut-off scores) when the pointing was performed during prism adaptation. The finding of an improvement after neutral pointing seems surprising, but can be easily explained in light of a form of visuomotor training embedded in the neutral pointing procedure. Pointing itself requires the patient to plan and perform a series of movements toward a visual stimulus placed in different positions of space. Manual pointing relies on a form of visuomotor coordination between the hand and eye (Henriques, Medendorp, Khan, & Crawford, 2002; see Crawford, Medendorp, & Marotta, 2004 and Buneo & Andersen, 2006 for reviews). Thus, the neutral pointing procedure trains the patient to orient the sensorimotor system toward the left side of space when a visual stimulus is presented there. If the orienting behaviour is reinforced by repetition of the procedure, the neutral pointing training might result in a long-term amelioration of neglect.

Prism adaptation can strengthen this type of visuomotor exercise, tapping into a similar synergy between the hand and eye. In addition to neutral pointing, prism adaptation also induces a systematic shift of visuomotor coordinates toward the left. Prismatic lenses indeed create a shift of the visual field toward the right side, as demonstrated by an initial *rightward* error in pointing to the visual target. The error signal is codified in visual eye-centred coordinates as the distance between the finger and the target in terms of visual angle. If patients are given visual feedback, after a few trials they will make a corrective movement to the target to compensate for this error and will progressively modify hand movement plans to reduce the target-finger gap. In this way, visual hand-centred coordinate systems may be reset by subtracting the visual error signal from the coordinates of the actual target signal (Redding & Wallace, 1993). As a consequence, the initial error in the visuomotor behaviour is corrected through visuomotor adaptation, and the visuomotor system resets toward the left. Since there is evidence that, during pointing, eye movements are yoked to hand movements and vice versa (Neggers & Bekkering, 2000; Prablanc, Echallier, Komilis, & Jeannerod, 1979; van Donkelaar,

1997), it has been hypothesized that under prism exposure, due to eye–hand coordination, the leftward deviation of hand movements also induces a leftward deviation of the oculomotor system (Angeli, Meneghello, Mattioli, & Ladavas, 2004; Ferber, Danckert, Joanisse, Goltz, & Goodale, 2003) and a consequent shift of visual attention toward the left side of the visual field, thus mediating the recovery of visual neglect. This hypothesis has been supported by showing an increase in the amplitude of the first leftward saccade after prism adaptation (Angeli et al., 2004) and a correlation between the correction of the pointing error during prism adaptation (i.e., Error reduction), the first saccade deviation and the amelioration of neglect obtained after the treatment (Serino et al., 2006).

The hypothesis that neglect recovery by prism adaptation is mediated by the oculomotor system was raised by those authors (Angeli et al., 2004; Frassinetti et al., 2002; Serino et al., 2006, 2009) who used a prism adaptation procedure with terminal exposure, i.e., very late visual feedback in target pointing. When the pointing movements are hidden until the final part of the movement when the index finger emerges, the pointing error during prism exposure is usually reliably high and the target-finger gap can be reduced only after a consistent amount of practice. Instead, in the concurrent (or early) exposure condition, i.e., when the limb is visible throughout the latter half of the pointing movement, the final pointing error is small and the target-finger gap can be reduced in the very first trials of practice. Because of the different visual feedback provided during pointing, terminal and concurrent exposure procedures vary for the coordinates system principally involved in the adaptation mechanisms, being primarily visual for the terminal exposure procedure and primarily proprioceptive for the concurrent exposure procedure (Redding, Rossetti, & Wallace, 2005). As a consequence, the terminal exposure procedure might produce a greater systematic leftward deviation of hand- and eye-centred reference frames and therefore a stronger effect on the oculomotor system and a stronger amelioration of neglect than the concurrent exposure procedure. Most of the studies on neglect recovery by prism adaptation have employed a concurrent exposure procedure (Berberovic, Pisella, Morris, & Mattingley, 2004; Farne et al., 2002; Ferber et al., 2003; McIntosh, Rossetti, & Milner, 2002; Morris et al., 2004; Nys et al., 2008; Rode et al., 2001; Rossetti et al., 1998; Rousseaux et al., 2006), whereas a minority of studies have used a terminal exposure procedure (Angeli et al., 2004; Frassinetti et al., 2002; Serino et al., 2006, 2007, 2009; Vangkilde & Habekost, 2010). Remarkably, in the latter studies, a long lasting amelioration of neglect was seen. Thus, it is possible to hypothesize that the different mechanisms of adaptation to prisms promoted by the two procedures are responsible for the different effects of treatment of neglect amelioration (see also Redding & Wallace, 2006; Striemer & Danckert, 2010).

Thus, the critical question is not simply whether remediation occurs after prism adaptation, but whether the leftward recalibration induced by a terminal exposure procedure is greater and more effective in ameliorating neglect than that induced by a concurrent exposure procedure. To this aim, in the present study, we directly compare the effects of two prism adaptation treatments, one based on terminal exposure procedure and the other based on concurrent exposure procedure. The results were compared with that of a treatment based on pointing with neutral goggles, in order to distinguish the specific effects of prism adaptation from those of a general visuomotor training provided by neutral pointing.

Thirty neglect patients were pseudo-randomly divided into 3 groups and assigned to terminal prism adaptation (TPA), concurrent prism adaptation (CPA), or neutral pointing (NP) procedures. All the treatments consisted of 10 daily sessions (5 sessions per week). Each session comprised 90 pointing movements toward a visual target presented in a variety of positions on the right, left, and at the centre of the visual field. Throughout the sessions, patients in

the TPA and CPA groups wore prismatic goggles deviating the visual field 10° toward the right, whereas patients in the NP group wore neutral goggles. Neglect was assessed at the time of admission and again after the treatment.

The following predictions were tested. If repeated visuomotor training is useful per se in reorienting the attentional systems to the left, an improvement of neglect should be found after NP, TPA and CPA treatments. If the leftward recalibration of sensory-motor coordinates due to prism adaptation specifically affects spatial representations, neglect amelioration found after the TPA and CPA treatments should be higher than that found after NP. Moreover, if the TPA procedure enhances the mechanism of leftward recalibration of eye–hand coordinates, neglect improvement should be higher after TPA than after CPA treatment.

2. Methods

2.1. Participants

Thirty right-brain-damaged patients with sub-acute (at least 2 months from the onset of illness) or chronic (at least 3 months from the onset of illness) left hemispatial neglect were recruited to participate in the study at the Istituto di Riabilitazione Santo Stefano, MC, Italy, the Clinica di Neuroriabilitazione, Azienda Ospedali Riuniti Lancisi-Salesi-Umberto I, Ancona, Italy and at the Centro Studi e Ricerche di Neuroscienze Cognitive, Cesena, Italy. They gave their informed consent to participate in the study. The study was conducted according to the Declaration of Helsinki (BJM 1991; 302; 1194), after approval from the Ethical Committee of the Istituto di Riabilitazione Santo Stefano. Patients were selected for the presence of hemispatial neglect after brain damage; the inclusion criterion was their pathological performance on the Behavioural Inattention Test battery (BIT, Wilson, Cockburn, & Halligan, 1987). Exclusion criteria were the presence of widespread mental deterioration (Mini-Mental State Examination score < 20) or psychiatric disorders. Ten patients were assigned to each group pseudo-randomly, with the last 6 patients distributed to match neglect severity between the 3 groups at the time of admission. Patients in each group did not differ in age (mean age: TPA=65 years, CPA=64; NP=61; $p > .27$ for all comparisons), education (mean years of schooling: TPA=9; CPA=7; NP=9; $p > .29$) or time after injury (mean months from injury, PTA=10; CPA=7; NP=15; $p > .19$). Patients were unaware of their group assignment; all were told that they had been enrolled in rehabilitation treatment for their spatial attention deficits. There were no dropouts. Patients' clinical and demographic data are presented in Table 1.

2.2. Neglect assessment

All patients underwent a standardized battery of tests for visuo-spatial deficits, the BIT (Wilson et al., 1987). BIT is composed of two scales, consisting respectively of conventional and behavioural tests. The Conventional scale includes **cancellation tasks, figure and shape copying, line bisection** and drawing from memory. The Behavioural scale includes tests which simulate different aspects of daily life activities, such as scanning a picture, dialling the telephone, reading a menu or an article, telling and setting the time, sorting coins, copying addresses and sentences, map navigation and sorting cards. The cut-off scores of the Conventional and Behavioural scale are 129 (range 0–146) and 67 (range 0–81), respectively. Patients were classified as having neglect when their score was below the cut-off score in at least one scale. The evaluation of neglect was performed in two different sessions. All tests were administered at the time of admission to the study, before any treatment (the baseline session), and again one week after the end of the treatment (post-treatment session). Testing sessions required approximately 3 h per patient distributed over 2 days to minimize fatigue.

For each patient, neglect assessment was performed by a neuropsychologist who did not administer the treatment to the same patient and who was unaware of the treatment procedure assigned to that patient.

2.3. Assessment of eye movements

TPA and CPA groups underwent eye movement recording during a reading task in two different sessions, performed before and after the treatments. To avoid an unnecessary fatigue, the NP group was not submitted to this test, as required by the Ethical Committee.

2.3.1. Apparatus

Subjects were seated in a dimly lit room with their heads stabilized straight ahead by means of an adjustable forehead and chin rest. The stimuli for the reading task were generated using custom software and displayed on a 15 in. colour monitor. The video screen was centred on the midsagittal plane of the subject's head and was viewed binocularly from a distance of approximately 42 cm. Horizontal eye movements were monitored using an infrared corneal reflection oculometer (Dr.

Table 1

Patients clinical and demographical data. The lesion site column reports the cortical and sub-cortical structures involved by the lesion (F: frontal; T: temporal; P: parietal; O: occipital; SC: sub cortical lesions).

Case	Gender	Age	Years of schooling	Months from illness	Site of the lesion
CPA1	F	64	7	3	T-P
CPA2	M	68	5	26	F-T
CPA3	M	67	5	2	T-P; SC
CPA4	M	59	5	7	F-P
CPA5	M	65	5	19	F-T
CPA6	M	63	13	30	F-T-P
CPA7	M	63	5	2	T
CPA8	M	61	13	3	T-P; SC
CPA9	M	61	5	2	T-P-O; SC
CPA10	M	64	5	3	T-P-O; SC
TPA1	M	61	23	3	F-P; SC
TPA2	M	54	13	5	T-P-O; SC
TPA3	M	65	8	2	T; SC
TPA4	M	75	8	11	F-T
TPA5	F	73	3	8	T-P
TPA6	F	58	8	2	P-SC
TPA7	M	57	13	2	F-T
TPA8	M	65	3	2	T-P; SC
TPA9	F	70	3	27	F; SC
TPA10	M	76	5	3	T; SC
NP 1	F	74	5	6	F-T
NP 2	F	63	5	2	T-P; SC
NP 3	M	53	8	7	F-T; SC
NP 4	F	61	13	5	F-T-P; SC
NP 5	M	60	5	11	T-P
NP 6	M	49	8	2	T
NP 7	M	52	18	4	F-T-P; SC
NP 8	F	72	7	5	T-P
NP 9	M	76	5	3	T; SC
NP 10	M	48	13	4	T-P; SC

Bouis Instruments, Germany) positioned in front of the subject's left eye. The eye movement tracker had high spatial resolution (about 5 min of arc) and its output was linearly related to eye position up to approximately 19.3° of visual angle (both horizontally and vertically). The analogue eye movement signals were sampled at 500 Hz, digitised by a lab-driver interface and stored on a hard disk for off-line analysis.

2.3.2. Material

Stimuli comprised 48 letter strings, 24 of 9 and 24 of 11 letters in length. Half of the stimuli ($n=24$) were common Italian words, and the remaining half ($n=24$) were non-words generated by changing two letters of each word. Word and non-word stimuli were presented in four separate blocks of 12 trials each (see Serino et al., 2006, for more details).

2.3.3. Reading task

A fixation cross was presented in the centre of the video screen. When the subject appeared to be correctly fixating the stimulus, the experimenter pressed a button to initiate the display. Then the central cross was extinguished and, after 100 ms, the stimulus was displayed for a maximum of 4000 ms. Patients had to look at the string and to report verbally what they read. Eye position recording started 100 ms before stimulus presentation and continued until the string went off. Patients' accuracy in the reading task was calculated as a proportion of strings correctly produced in the available time.

2.3.4. Eye movement analysis

As an index of eye movement performance, we measured the first saccade landing location, which provides a measure of the first saccade amplitude. A detailed description of this analysis is provided in Serino et al. (2007).

2.4. Treatment procedure

2.4.1. Prism adaptation procedure

Patients were seated at a table. In front of them there was a wooden box (height 20 cm, depth 34 cm at the centre and 18 cm at the periphery, width 72 cm). The box was open on the side facing the patient and on the opposite side, facing the experimenter. A visual target (a pen) was presented manually by the experimenter at the distal edge of the top face of the box. Visual targets were presented 90 times 60 cm away from the patient's midline subtending a total visual space of 50°. Thirty times, the visual targets were presented in a variety of positions in the patient's right visual field, 30 times in the left and 30 times in the centre, in random order. The

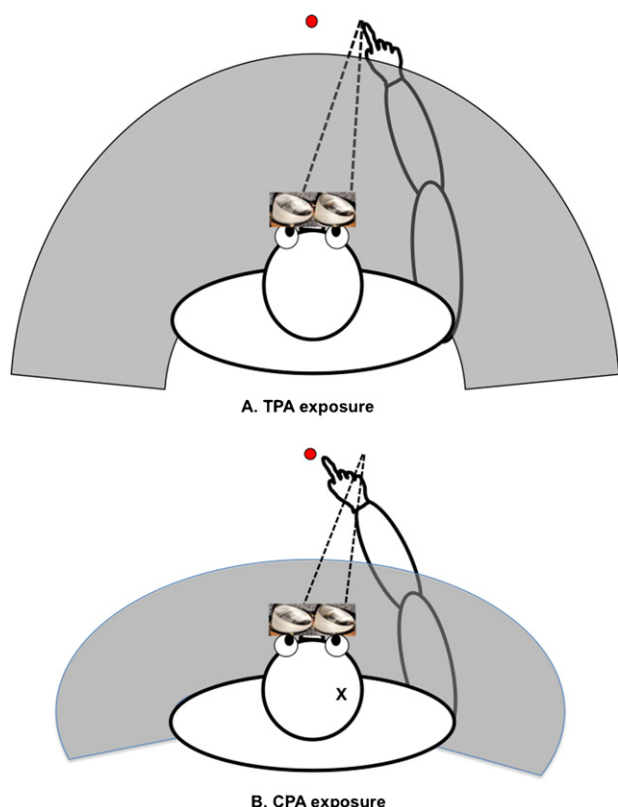


Fig. 1. TPA and CPA treatments apparatus. A schematic view of a session of pointing under prism exposure. (A) The TPA procedure, when only the final part of the pointing movement was visible to the patient. (B) The CPA procedure, when half of the pointing movement was visible.

experimenter recorded patients' pointing spatial accuracy as the distance between the target position and the final position of the patient's finger. A graduated transparent barrier, invisible to the patient, was used to assess pointing deviation (see below). Patients were asked to keep their right (ipsilesional) hand on their chest, at the level of the sternum (hand starting position), and to point with the index finger toward the pen, at a fast but comfortable speed. The pointing arm movement was executed within the wooden box whose upper side was moved forward and backward. This enabled the visual feedback to vary in the three different conditions (visible pointing, terminal phase of the trajectory; visible pointing, half movement trajectory; invisible pointing, see below). Patients underwent the treatment in 10 sessions, one per day, which took about 30 min each, over a period of 2 weeks. The pointing task was performed in three experimental conditions: pre-exposure (visible and invisible pointing), exposure (visible pointing) and post-exposure (invisible pointing).

2.4.2. Pre-exposure condition

Patients were required to point toward 60 targets randomly presented in a variety of positions in the right visual field, left visual field, and at the centre. Patients performed half of the pre-exposure trials with "visible" pointing (for TPA patients, with the terminal phase of the movement visible), and for CPA patients, with half of the movement visible), which became the baseline for the exposure condition, and half with "invisible" pointing, which became the baseline for post-exposure condition (see below).

2.4.3. Exposure condition

Patients performed the same task wearing the prismatic (Optique Peter, Lyon) or the neutral goggles. The prismatic goggles were fitted with wide-field prismatic lenses inducing a 10° shift of the visual field to the right. Patients were asked to point rapidly to 90 targets presented in a random order (30 targets at the centre, 30 on the right and 30 on the left). During the exposure condition, for the TPA group, the pointing movement was hidden below the top face of the box until the final part of the movement (approximately the last 12 cm) when the index finger emerged beyond the distal edge of the box. For the CPA group, the entire latter half of the pointing movement was visible (see Fig. 1). For the NP group, the treatment was the same as for the TPA group, except that patients performed their pointing exercises while wearing neutral goggles.

2.4.4. Post-exposure condition

Immediately after prism removal, patients were required to point toward 30 targets (10 at the centre, 10 on the right and 10 on the left). The pointing movement was performed entirely below the top face of the box, so that the movement was not visible at any stage (invisible pointing).

2.4.5. Visuomotor indices of prism adaptation

To ensure that patients treated with prisms actually achieved a leftward correction of their visuomotor behaviour, two indices of sensorimotor effects were measured: *Error reduction* and *After effect*. *Error reduction* refers to the online correction of pointing movements during prism adaptation, whereas *After effect* refers to the amount of leftward deviation in pointing achieved after prism adaptation. These 2 indices are considered the hallmark of adaptation to prisms (Redding & Wallace, 2006).

Patients' pointing errors were recorded by a therapist who is an expert in PA procedure and who sat in front of the patient. A transparent Plexiglas barrier, etched with a measurement scale for easy assessment, was placed between the therapist and the patient. Hidden from the patient's view, the Plexiglas barrier allowed the therapist to assess where exactly the patient pointed during the task. Pointing was made with the index finger, with pointing error being calculated as the deviation made between the final pointing position and that of the visual target, measured in degrees of visual angle. Rightward deviation was coded as positive and leftward deviation was coded as negative. To assess *Error reduction*, pointing error was measured in the pre-exposure condition (with visible pointing) as a baseline, and then during each session of treatment (exposure condition). To assess *After effect*, pointing error was measured in the pre-exposure condition (with invisible pointing) and after each session of treatment (post-exposure condition).

3. Results

3.1. Visuomotor effects of prism adaptation

3.1.1. Error reduction effect

In order to demonstrate that patients gradually adapted to the optical displacement induced by the prisms (i.e., *Error reduction effect*), mean pointing errors during each session of treatment were computed and shown in Fig. 2A. In order to test whether the two different prism adaptation conditions induced different levels of *Error reduction*, the mean pointing error was compared between TPA and CPA patients by means of a mixed ANOVA with the within-subjects factor *Session* (baseline and 10 sessions of treatment) and the between-subjects factor *Group* (TPA and CPA). The main effects of *Session* [$F(10,180)=6.31$; $p<.001$] were significant. Patients demonstrated no pointing error in the baseline session (mean pointing error = $-.002$; s.e.m. = $.02$), whereas their pointing was deviated in the direction of the prismatic shift beginning with the first session of treatment (mean pointing error = $.41$; s.e.m. = $.10$). The main effect of *Group* [$F(1,18)=9.62$; $p<.01$] was also significant. CPA patients were more accurate in pointing during prism exposure (mean pointing error for 10 sessions of treatment = $.08^\circ$ of visual angle; s.e.m. = $.01$) than TPA patients (mean = $.29$; s.e.m. = $.05$). Finally, the interaction *Group* $\times$ *Session* was significant [$F(10,180)=1.89$; $p<.05$]: while pointing errors did not differ between the two groups in the baseline session without prisms ($p=.41$), TPA patients showed larger pointing errors than CPA patients during prism exposure. TPA patients showed initially much higher pointing errors than CPA patients, and then progressively compensated through the treatment. Indeed, while in the first session of treatment, pointing errors were significantly higher in TPA (mean = $.58$; s.e.m. = $.15$) than in CPA patients (mean = $.18$; s.e.m. = $.01$; $p<.05$), in the final session of treatment, pointing errors were no longer significantly different between the two groups (TPA: mean = $.17$; s.e.m. = $.03$; CPA: mean = $.07$; s.e.m. = $.02$; $p=.13$) (see Fig. 2A).

It is normally accepted that *Error reduction* occurs progressively along the successive trials of prism adaptation. In the initial trials, the pointing is more strongly displaced in the direction of the optical shift, and thus pointing errors is higher, whereas after a sufficient amount of trials, no pointing error is evident as a result of visuomotor adaptation. This is especially true for the

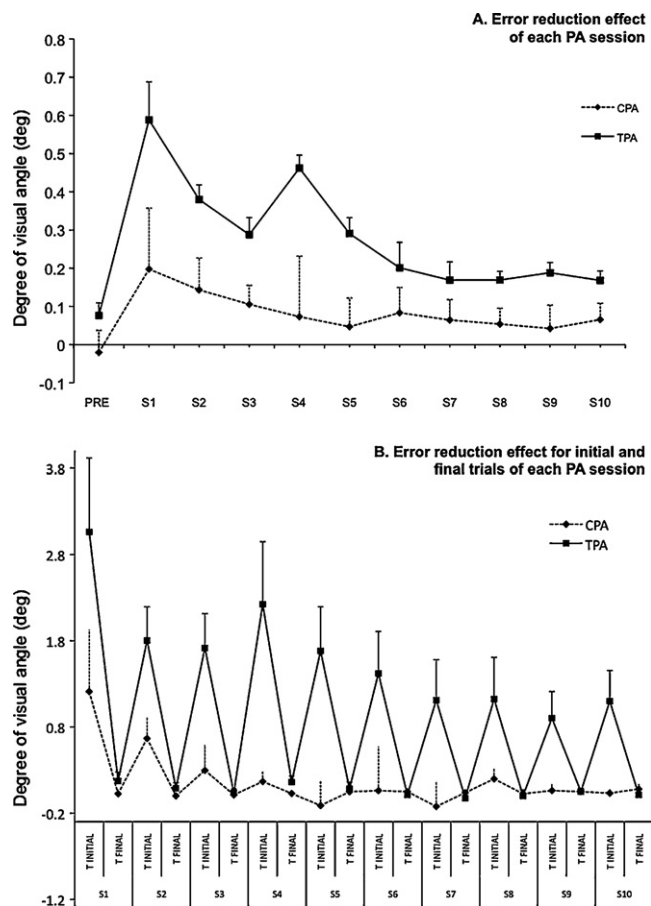


Fig. 2. Error reduction effect of prism adaptation. (A) Pointing errors recorded during visible pointing task before prism adaptation (PRE) and during each prism adaptation session (S1–S10) for TPA and CPA patients. (B) Pointing errors for TPA and CPA patients in the initial (trials 1–9) and the final trials of prism adaptation (trials 81–90) of each prism adaptation session.

TPA procedure in comparison to the CPA procedure (Redding & Wallace, 2006). To test for this prediction in the present sample of patients, we performed an additional analysis on pointing errors during prism exposure. For each treatment session we compared the mean pointing errors for the initial 9 trials (1–9) with that of the final 9 trials (81–90) of prism adaptation (3 leftward, 3 centre, 3 rightward), in both TPA and CPA patients. ANOVA was conducted, with within-subjects factors as 'Session' (10 treatment sessions in total), and 'Trials' (initial and final 9 trials), and the between-subjects factors as 'Group' (TPA and CPA). The main effects of Session [$F(9,162)=6.26$; $p<.00001$], Trial [$F(1,18)=16.28$; $p<.001$] and Group [$F(1,18)=9.39$; $p<.01$] were significant. As shown in Fig. 2B, pointing errors diminished progressively over the 10 treatment sessions (e.g. mean pointing errors in the first session = 1.11° ; s.e.m. = $.26$; in the last session = $.31$; s.e.m. = $.08$; $p<.0001$, Duncan corrected), and between the first and last 9 trials at each session (e.g. first 9 trials, mean pointing error = $.93$ d.e.g.; s.e.m. = $.22$; last 9 trials, mean pointing error = $.04$ d.e.g.; s.e.m. = $.01$). In addition, the pointing errors were higher in the TPA group (mean pointing error = $.83$ d.e.g.; s.e.m. = $.15$) than in the CPA group (mean pointing error = $.15$ d.e.g.; s.e.m. = $.15$). More importantly, the between groups difference for pointing errors is attributable to a different pattern of error reduction effect throughout the PA trials, as shown by a significant Group $\times$ Trial interaction [$F(1,42)=8.78$; $p<.01$]. As Fig. 2B shows, in the TPA group pointing error was significantly higher during the initial trials (mean = 1.59 ; s.e.m. = $.37$) than in the final trials of each session (mean = $.03$; s.e.m. = $.02$; $p<.0001$;

Duncan corrected), whereas in the CPA group the same degree of low pointing error (close to 0°) was recorded during the initial trials (mean = $.27$; s.e.m. = $.23$) as well as in the final trials of each session (mean = $.03$; s.e.m. = $.01$; $p=.73$). In other words, at the beginning of each treatment session TPA patients made significantly more pointing errors than CPA patients ($p<.0001$; Duncan corrected), whereas at the end of each treatment session patients from both groups compensated in the same way for the optical displacement as induced by the prism goggles ($p=.93$). Finally, the Session $\times$ Trial type interaction was also significant [$F(9,21)=5.83$; $p<.0001$], showing that the difference in pointing errors between initial and final trials at each treatment session was higher in the first sessions, progressively diminishing toward the last sessions (e.g. first treatment session: initial trials, mean pointing error = 2.13 ; s.e.m. = $.52$; final trials, mean pointing error = $.09$; s.e.m. = $.05$; $p<.00001$, Duncan corrected; last treatment session: initial trials, mean pointing error = $.59$; s.e.m. = $.17$; final trials, mean pointing error = $.04$; s.e.m. = $.04$; $p=.13$).

Error reduction data cannot be statistically analyzed for the NP patients since pointing errors were virtually 0 in all cases; thus no variance could be computed.

3.1.2. After effect

In order to demonstrate the presence of an After effect induced by prism adaptation, and in order to study whether such an effect varied for the TPA and CPA treatments, pointing errors recorded in before the treatment (baseline) and after each session of prism adaptation were compared by means of an ANOVA with the within-subjects factor of Session (baseline and 10 post adaptation sessions) and the between-subjects factor Group. Only the main effect of Session was significant [$F(10,180)=23.71$; $p<.0001$]. As Fig. 3 shows, while no pointing error was recorded during the baseline session (mean = $-.37$; s.e.m. = $.44$), a significant deviation of pointing toward the left was recorded after each session of treatment (mean pointing error for the 10 prism adaptation sessions = -6°). This pointing behaviour was not different between the two groups of patients, since neither the main effect of Group ($p=.93$) nor the Group $\times$ Session interaction ($p=.84$) was significant. After effect scores cannot be statistically analyzed for the NP group, since pointing errors were always close to 0.

In summary, the two types of prism adaptation procedures induced different Error reduction effects, being more demanding for TPA patients to adapt to prism than for CPA patients. However, the total amount of After effect induced by the treatment was comparable between the two procedures.

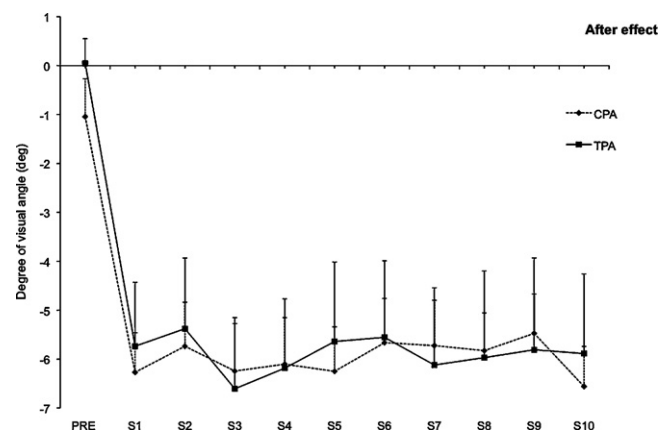


Fig. 3. After effect of prism adaptation. Pointing errors recorded during invisible pointing task before prism adaptation (PRE) and after each session of prism adaptation (S1–S10) for TPA and CPA patients.

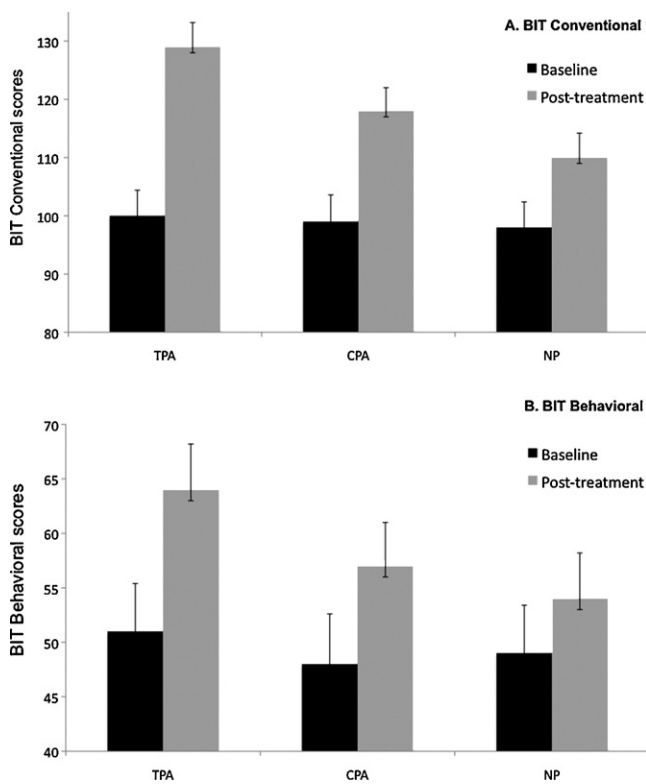


Fig. 4. Visuo-spatial effect of prism adaptation. (A) BIT Conventional scores before and after treatment for TPA, CPA and NP patients. (B) BIT Behavioural scores before and after treatment for TPA, CPA and NP patients.

3.1.2.1. Neglect recovery in TPA, CPA and NP groups. In order to study the effect of the different types of treatment on neglect recovery, pre and post treatment neuropsychological assessments were analyzed: two ANOVAs were conducted on BIT-Conventional and BIT-Behavioural scores with the within-subjects factor 'Session' (baseline and post-treatment sessions) and the between-subjects factor 'Group' (TPA, CPA and NP). Duncan post hoc comparisons were conducted, when necessary. As far as the BIT-Conventional scores are concerned (see Fig. 4A), the main effect of Session ($F[1,27]=46.38$; $p<.0001$) was significant. Patients' scores were higher after the treatment (mean score = 99; s.e.m. = 3.7) than at baseline (mean = 119; s.e.m. = 3.6). The significant Group $\times$ Session interaction ($F[2,27]=3.39$; $p<.05$) better clarifies this outcome. Scores by TPA patients strongly improved between pre-treatment (mean = 100; s.e.m. = 6.5) and post-treatment assessment (mean = 129; s.e.m. = 6.2; $p<.0001$). When comparing BIT-Conventional scores before and after the treatment, a moderate improvement was also found in the CPA group (baseline mean score 99; s.e.m. = 6.5; post-treatment mean score = 118; s.e.m. = 6.2; $p<.001$) and in the NP group (baseline mean score = 98; s.e.m. = 6.5; post-treatment mean score = 110; s.e.m. = 6.2; $p<.05$). Importantly, although at baseline, BIT-Conventional scores of TPA patients did not differ from those of CPA ($p=.78$) and NP patients ($p=.81$), and BIT-Conventional post-treatment scores were significantly higher in TPA patients than in CPA ($p<.01$) and NP ($p<.001$; see Fig. 4A) patients, with the difference between the last two conditions marginally significant ($p<.1$).

Similar results were obtained for the BIT-Behavioural Scores (see Fig. 4B). The main effect for Session ($F[1,27]=46.93$; $p<.0001$) was significant. Patients' scores were higher after the treatment (mean score = 58; s.e.m. = 2.6) than at baseline (mean = 50; s.e.m. = 2.3). The two-way Group $\times$ Session interaction ($F[2,27]=4.01$; $p<.05$) was also significant. Scores by TPA patients

strongly improved between pre-treatment (mean = 51; s.e.m. = 4.4) and post-treatment assessment (mean = 64; s.e.m. = 4.2; $p<.0001$). A moderate improvement was also found in the CPA group (baseline mean score 48; s.e.m. = 4.6; post-treatment mean score = 57; s.e.m. = 4; $p<.01$) and in the NP group (baseline mean score = 49; s.e.m. = 4.4; post-treatment mean score = 54; s.e.m. = 4.2; $p=.06$). In the baseline condition, BIT-Behavioural scores were not significantly different in TPA patients as compared to CPA ($p=.31$) and NP patients ($p=.74$). BIT-Behavioural scores after PA treatment were significantly higher in TPA patients compared to CPA ($p<.01$) and NP ($p<.001$) patients, with the difference between the last two conditions being marginally significant ($p=.1$).

Furthermore, considering the combined outcome of the BIT Conventional and Behavioural scores, while at the baseline, patients from the three groups were significantly below the cut-off score, after-treatment BIT scores of CPA patients, were marginally below the cut-off [$t(1,9)=-2.04$; $p=.07$], with 4 out of 10 patients reaching the cut-off. In contrast, in the TPA group, post-treatment BIT scores were not different from the cut-off [$t(1,9)=-.75$; $p=.44$]. Six out of 10 of these patients reached the cut-off after the TPA treatment. Scores for patients in the NP group remained significantly below the cut-off score after the treatment [$t(1,9)=2.58$; $p<.05$]. Only 1 out of 10 patients from the NP group obtained a score above the cut-off after the treatment.

An ANOVA with Group (TPA and CPA) and Session (baseline and post-treatment) was also conducted on the percentage of correctly read letter strings. Only the main effect of Session was significant [$F=10.73$; $p<.01$]. Patients were significantly more accurate in reading after the treatment (correct reading = 60%; s.e.m. = 6) than before (40%, s.e.m. = 7). The main effects of the Group and the Group $\times$ Session interactions were not significant ($p=.58$ and $.59$, respectively). In order to control for the great variability in patients' reading accuracy before treatment, an index of improvement was calculated as the difference between baseline and post-treatment scores divided by the difference between maximal accuracy (100%) and baseline scores. The reading accuracy index was compared between the CPA and TPA groups by means of an independent sample t -test. Although numerically the improvement was higher for the TPA group (37%; s.e.m. = 26) than for the CPA group (20%; s.e.m. = 39%), this difference was not statistically significant ($p=.29$).

3.1.3. Effects of TPA and CPA on eye movements

To verify that TPA and CPA induced a different leftward deviation of eye movements, the amplitude of the first saccade recorded in the baseline and post-treatment sessions was compared in the two groups of patients. Because of technical problems, eye movement data were available for only 9 of the 10 patients from each group.

An ANOVA was performed on the first saccadic endpoint, taking Group (TPA and CPA) as the between-subjects factor and Session (baseline and post-treatment) as the within-subjects factor. The main effect of Session was significant [$F(1,16)=24.01$; $p<.0001$]. The first saccadic end point fell significantly more leftward after treatment (-1.27° of visual angle; s.e.m. = .31) than before treatment ($-.76^\circ$ of visual angle; s.e.m. = .30). However, the effect of prism adaptation was different between the two groups, as shown by the significant interaction Group $\times$ Session [$F(1,16)=3.45$; $p<.05$]. In the baseline condition, the first saccadic endpoint fell close to the middle of the letter string (TPA mean deviation: -0.42 , s.e.m. = .30; CPA mean deviation: -0.47 ; s.e.m. = .31), without a significant difference between the two groups ($p=.95$). In contrast, in the post-treatment session, the first saccade was displaced more toward the left in the TPA group (-2.08° ; s.e.m. = .3) compared to the CPA group (-1.09° ; s.e.m. = .4; $p<.05$) (see Fig. 5). The main effect of Group was not significant ($p=.26$).

3.1.4. Relationship between prism adaptation procedure and visuomotor, oculomotor and visuospatial effects of prism adaptation

The previous results show that TPA treatment compared to CPA treatment was characterized by: (a) a more prominent Error reduction effect; (b) a greater improvement of visuospatial abilities; (c) a stronger leftward deviation of eye movements. These findings can be explained by considering that TPA treatment emphasizes a strategic correction of visual coordinates, mainly affected in visuospatial neglect, whereas the CPA treatment emphasizes a more automatic adaptation of proprioceptive coordinates. We hypothesize that the strategic correction of visual coordinates during TPA treatment is the basis of both the amelioration of neglect and the deviation of the eye movement system after the treatment. If this is the case, the amount of Error reduction effect in the first phase of the treatment should predict the improvement in BIT scores and the leftward displacement of the first saccade obtained at the end of the treatment in TPA patients. These relationships should not be expected in CPA patients.

To test these hypotheses, for each patient, we calculated an index of Error reduction effect for the first week of treatment by averaging pointing errors for the first 5 sessions of treatment (see Serino et al., 2006) and an index of neglect amelioration, computed as the difference between the post-treatment BIT score and the baseline BIT score, divided by the difference between the maximum BIT score (114) and the BIT score at the baseline. This index indicates the percentage of the improvement compared to the maximum possible improvement obtainable. Finally, an index of eye movement response was calculated as the difference between the first saccadic endpoint post-treatment minus the first saccadic endpoint at the baseline.

The linear regression between the Error reduction effect for the first week of treatment and the index of neglect amelioration was significant for TPA patients ($R^2 = .43$; $F = 7.18$; $p < .05$; $\beta = -.46$), but not for CPA patients ($R^2 = .19$; $F = .16$; $p = .70$; $\beta = -.21$). Thus, patients who exhibited a greater correction of the initial pointing error through the first week of TPA treatment also showed a stronger improvement of neglect (see Fig. 6A). The relationship between Error reduction and neglect improvement was absent in patients exposed to the CPA treatment.

Analogously, the linear regression between the Error reduction in the first week of treatment and the displacement of the first saccade was significant for TPA patients ($R^2 = .58$; $F = 8.60$; $p < .05$; $\beta = -3.24$), but not for CPA patients ($R^2 = -.10$; $F = .79$; $p = .40$; $\beta = -1.80$). As Fig. 6B shows, patients showing a stronger Error reduction effect during the first week of TPA treatment also

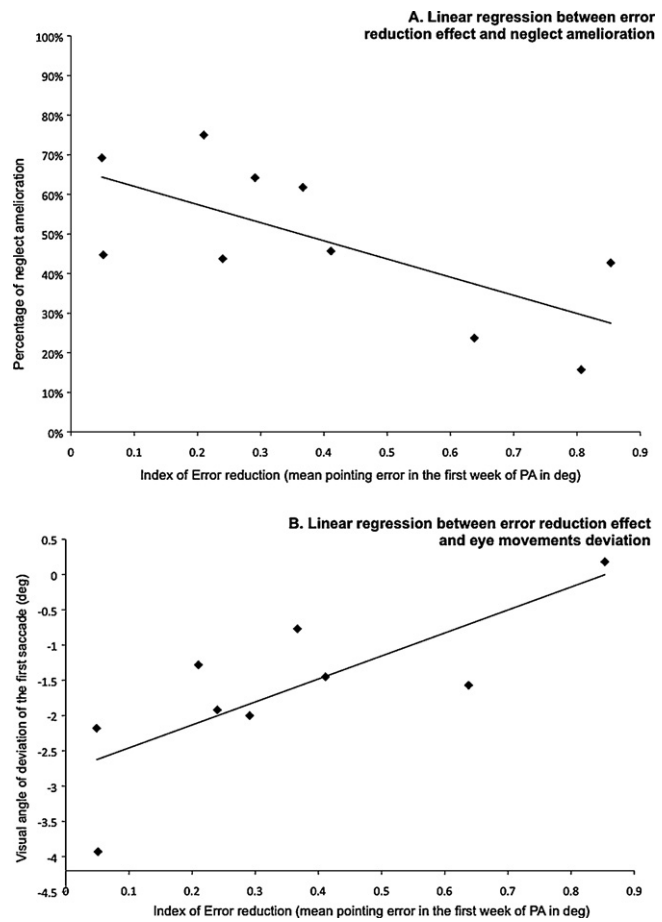


Fig. 6. Relationship between Error reduction effect and visuo-spatial and oculomotor effects of TPA. (A) The linear regression between the level of Error reduction recorded in the first week of prism adaptation and the amount of amelioration of neglect obtained after the treatment in TPA patients. (B) The linear regression between the level of Error reduction in the first week of prism adaptation and the leftward deviation of the first saccade obtained after the treatment in TPA patients.

obtained a stronger leftward deviation of the first saccadic endpoint at the end of the treatment. This was not the case for CPA patients.

After-effect was not statistically related to neglect improvement: the index of after-effect for the first week of treatment did not correlate with the index of neglect amelioration in TPA and CPA patients (all $p_s > .33$).

4. Discussion

In this study we compared the effects of two types of prism adaptation treatments, based on two different prism adaptation procedures, namely TPA (i.e., a terminal exposure with very late visual feedback in target pointing) and CPA (i.e., a concurrent, or early, exposure, with the limb continuously visible during the latter half of the pointing movement) and those of 2 weeks of visuomotor training performed without prismatic goggles (NP). The results show that the repetition of pointing movements toward visual stimuli improved visuospatial performance in patients treated with prism adaptation and with NP. However, neglect amelioration was significantly greater when the pointing was performed under the influence of prisms. Furthermore, the improvement was greater in patients treated with the TPA procedure compared to patients treated with the CPA procedure.

These findings confirm previous results showing that the neglect improvement can be obtained by means of visuomotor training based on pointing tasks (Serino et al., 2009). This exercise requires

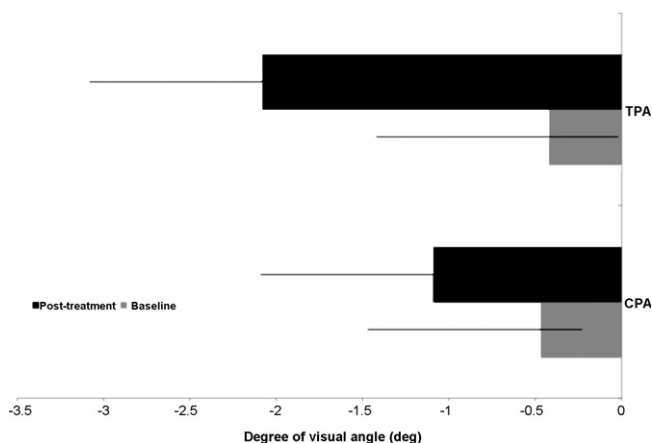


Fig. 5. Oculomotor effect of prism adaptation. The landing location of the first saccade recorded before and after the treatment for TPA and CPA patients.

the patient to plan and perform a series of movements toward a visual stimulus presented in different positions in the visual field. Since pointing relies on a form of visuomotor coordination between the hand and eye (Neggers & Bekkering, 2000) pointing exercises might train the patient to orient the eye–hand sensorimotor system toward the left side of space. This effect can be strongly reinforced by prism adaptation, as shown by the finding that neglect improvement was greater in patients exposed to prisms and that the amelioration of visuo-spatial disabilities was associated with a low-order reorganisation of visuomotor response system aimed at compensating the rightward optical displacement induced by prisms.

4.1. Low-order reorganisation of visuomotor response and neglect amelioration

The effects of prism adaptation treatments will be discussed in light of two dissociable processes thought to be involved in prism adaptation: a strategic control mechanism, involving the correction of errors in motor output, termed *strategic recalibration*, and a more automatic, sensorimotor, component involving a realignment of the visual and proprioceptive maps so as to produce spatially accurate behaviour, termed *adaptive spatial alignment* (Newport & Jackson, 2006; Redding & Wallace, 2006; Redding, Clark, & Wallace, 1985; Redding et al., 2005; Weiner, Hallett, & Funkenstein, 1983).

When subjects view a stimulus through rightward displacing optical prisms, their angle of gaze must be rightward of the veridical angle to that stimulus in order to fixate it. As a consequence, if subjects are asked to point to the visual stimulus, they initially produce a rightward error. In order to correct for the visual perturbation induced by prisms, they must, in effect, plan a reach direction that is incorrect: they must direct their limb to a target location other than that of the direction of their gaze. Thus, the motor program for guiding the limb must be decoupled from the oculocentric coordinate system in which the target location/angle of gaze is specified. Only then, the limb can be re-directed to the target location, specified by the final limb position, to the left of the visual target that corresponds to the veridical spatial location of the visually displaced target. This crucial step, i.e., that of directing the limb away from the naturally programmed target location, is at the basis of the *strategic recalibration mechanism* in prism adaptation, and is mainly responsible for the *Error reduction effect* during prism exposure (see Redding et al., 2005).

Although critical, this is not the only important effect in prism adaptation. Indeed, in order to produce spatially accurate behaviour, a re-alignment of the visual and proprioceptive maps needs to occur through the repetition of a number of pointing movements (Redding & Wallace, 1996; Weiner et al., 1983). Spatially congruent eye and arm information needs to be integrated to produce coordinated eye–hand movements (Battaglia-Mayer & Caminiti, 2002). *Adaptive spatial alignment* takes longer to develop, but it is here that the prism *After effect* is observed. The prism *After effect* following adaptive spatial alignment is relatively long lasting and persists despite conscious knowledge of the removal of the visual perturbation (Clower & Boussaoud, 2000). These two effects of prism adaptation in neglect patients will be discussed separately in the following sections.

4.1.1. Strategic recalibration in TPA and CPA (Error reduction)

When the prisms were applied to patients from the present sample, they made important pointing errors in the direction of the visual perturbation. However, the prism effect differed depending on the different prism adaptation procedures. The initial pointing error of TPA patients was about 30% of the total optical displacement produced by the prisms, whereas that of CPA patients was close to 10% of the total optical displacement (mean pointing

error for initial trials of the first session of treatment: TPA = 3°; CPA = 1.20°; see Fig. 3B). The difference in the size of the pointing error initially induced by the two procedures is explained by the fact that CPA patients were allowed to see half of their arm movement and in this way they could correct the motor program during the reaching phase of the movement. This early correction was not possible for TPA patients, who got visual feedback of their arm movement only when it was almost completed. As a result, the spatial discrepancy between the naturally occurring motor program (planned in coordinates relative to the perceived target location) and the actually produced, incorrect motor program (planned in hand-based coordinates relative to the real target location) was more prominent in TPA patients than in CPA patients (see also Redding & Wallace, 1993; Rossetti, Koga, & Mano, 1993). Because of the large terminal error, TPA patients had to direct their limb further to the left of the target location compared to CPA patients in order to correct for the spatial discrepancy between eye and hand coordinates.

Since eye and hand information are normally integrated in order to produce coordinated eye–hand movements (Battaglia-Mayer & Caminiti, 2002; Neggers & Bekkering, 2000; Prablanc et al., 1979; van Donkelaar, 1997), in TPA patients, the oculomotor system also was re-directed further to the left by prism adaptation. Indeed, in these patients the stronger Error reduction effect was associated with a greater leftward deviation of eye movements compared to CPA patients. Interestingly, in TPA patients, the level of Error reduction during the first week of treatment was correlated both with the leftward deviation of the first saccade and with neglect amelioration after the treatment: the more the patient corrected his/her pointing behaviour during the initial phases of prism exposure, the more leftward deviated his/her oculomotor system was at the end of the treatment and the greater was the neglect improvement. In contrast, in CPA patients, although a leftward deviation of the first saccade and some neglect amelioration did occur at the end of treatment, these effects were smaller than those in TPA patients and did not correlate with the Error reduction effect.

Thus, it seems that the degree of the visual error initially induced by prism adaptation is an important factor in determining the magnitude of the strategic recalibration process and the consequent leftward shift of the oculomotor system and neglect amelioration. In this respect, it seems that the TPA procedure is better than the CPA procedure in inducing a leftward deviation of eye movements and an amelioration of neglect. It is possible to speculate that the increase in the amplitude of the first leftward saccade after prism adaptation promotes a shift of visuo-spatial attention toward the left, thus mediating the recovery of visual neglect. The initial leftward deviation of eye movements induced by prism adaptation might be continuously reinforced by stimuli from the external environment and be further implemented in everyday life. Therefore patients can begin to interact normally again with the left spatial field of their environment, similar to before suffering brain damage. In this way, further reinforcement of the strategic recalibration mechanism, as induced by PA, will therefore afford the patient gradual access to a normal representation of space again. Being that the amplitude of the leftward saccade is larger in TPA than in CPA patients, it is not surprising to find a greater improvement of neglect in the former group compared to the latter.

4.1.2. Adaptive spatial alignment in TPA and CPA patients (After effect)

The measure that has been traditionally considered critical to demonstrate prism adaptation (Harris, 1974; Redding & Wallace, 1993; Welch, 1978) is the *After effect*, rather than the Error reduction effect. But, interestingly, *After effect* seems not to be directly related to neglect recovery, since this index was not significantly different in the TPA and CPA groups and did not correlate with

Table 2

Studies on the effectiveness of prism adaptation for neglect recovery, comparing an experimental and a control group.

Study	Patients (experimental + control group)	Time from injury	Treatment duration: sessions (over N weeks)	Prism adaptation procedure	Higher effects for experimental patients
Rossetti et al. (1998)	6 + 6	Not reported	1	CPA	Up to 2 h
Frassinetti et al. (2002)	7 + 6	>3 months	20 (2)	TPA	Up to 1 months
Serino et al. (2006)	16 + 8	>3 months	10 (2)	TPA	Up to 3 months
Rousseaux et al. (2006)	10 + 8	17–102 days	1	CPA	NO
Nys et al. (2008)	10 + 6	4–23 days	4	CPA	Immediate effects. NO at 1 month
Serino et al. (2009)	10 + 10	>1 month	10 (2)	TPA	Long lasting
Vangkilde and Habekost (2010)	6 + 5	>5 months	20 (2)	TPA	Long lasting
Turton et al. (2009)	18 + 16	≈40 days	14 (2)	?	NO

neglect amelioration nor with the leftward deviation of eye movements (see also Sarri, Kalra, Greenwood, & Driver, 2006; Serino et al., 2006, 2007). The explanation for these results could be found in the nature of the visuomotor correction during the After effect, which mostly reflects a reorganisation of the proprioceptive hand-centred coordinate system. Indeed, during the post-exposure test, the hand is no longer visible, thus there is no match between visual and proprioceptive hand information. In this case, the pointing bias after prism exposure reflects an acquired movement correction, expressed mainly, although not exclusively, in terms of proprioceptive hand-centred coordinates. This may be the reason why the After effect, as revealed with **open-loop manual pointing** to a visual target, does not correlate with the deviation of the oculomotor system, strictly linked to the visual system, and with the recovery of visuo-spatial neglect.

It is worth noting that other measures of After effect of prism adaptation can be considered. For instance, pointing toward the body midline ahead without vision (proprioceptive straight-ahead task) can be used to assess selectively the *proprioceptive* After effect of prism adaptation (as in Rossetti et al., 1998). On the other hand, stopping a moving spot as it crosses the perceived midline, without any limb movement (visual straight-ahead judgement task), can be used to measure selectively the *visual* After effect. Strategic recalibration – mainly underlying TPA – and adaptive spatial alignment – mainly underlying CPA – can induce stronger visual or proprioceptive After effects, respectively (see Redding & Wallace, 2010 for an exhaustive model). For this reason, it is possible that different measures of After effect (proprioceptive or visual) correlate with any form of neglect improvement, but at the moment, there are no data, at least to our knowledge, to support, or reject this possibility.

To summarize, the results concerning the relationships between the effects of prism adaptation on different sensory and motor representations suggest that the process of neglect improvement by prism adaptation, probably mediated by the reorganization of the oculomotor system, needs to be guided mainly by visual information. For this reason, a prism adaptation procedure that stresses the visuomotor eye–hand components of the adaptation process seems most effective for the rehabilitation of neglect.

5. Clinical implications

The present data are important in that they shed some light on the mechanism involved in the resetting of sensory-motor maps in neglect patients after the PA treatment. These insights might be clinically relevant in explaining some discrepancies between the different results reported by several rehabilitative studies using prism adaptation (see Table 2). While previous reports by our own group (Frassinetti et al., 2002; Serino et al., 2006, 2009) and by other authors (Vangkilde & Habekost, 2010; see also Shiraishi et al., 2008) repeatedly demonstrated the effectiveness of prism adaptation for long lasting neglect recovery, other authors have found no or negligible effects of prism adaptation treatment. Rousseaux et al. (2006) could not replicate immediate nor sustained (after 3 days) bene-

fits following one session of treatment. Ferber et al. (2003) found no functional relevance of changes in eye movements after prism adaptation, which supports the claim that prism adaptation does not improve the core attentional aspects of the neglect syndrome (see also Morris et al., 2004). Nys et al. (2008) found faster neglect recovery in patients treated with prism adaptation compared to patients treated with NP immediately after the treatment, but this difference disappeared one month later. Finally, Turton et al. (2009) found no difference in the improvement of neglect scores between two groups of acute neglect patients, one submitted to prism adaptation and the other to NP. However, there is a key difference in the treatment procedures used by studies showing a positive effect of prism adaptation on neglect recovery and studies showing no effect (see Table 2). Our group and Vangkilde and Habekost always used a TPA procedure, whereas Rousseaux et al. (2005), Ferber et al. (2003), Morris et al. (2004) and Nys et al. (2008) used a CPA procedure. Turton et al. (2009) state that they have used a procedure analogous to that described in Frassinetti et al. (2002). However, in that paper, Fig. 1, illustrating the treatment apparatus, shows that, although the terminal part of the pointing movement was visible as in a standard TPA procedure, patients could see a part of the initial phase of their pointing movement. This early visual feedback about the position of the arm renders this treatment procedure not comparable to the TPA procedure used in other studies (see Redding & Wallace, 2006).

Secondly, as one can see in Table 2, there are two other important confounding effects to be considered when comparing the results from different studies: the duration of the treatment and the time post-injury of the patients. First, it is not possible to directly compare the effects of 1 session of prism adaptation treatment, as in Rousseaux et al. (2006) or Ferber et al. (2003), or 4 sessions of treatment, as in Nys et al. (2008), with those of 2 weeks of treatment. A single session of prism adaptation can induce some amelioration of neglect, but then, the prism adaptation effects require enough repetition to be stably implemented and to produce a general and long-lasting benefit for the majority of patients. Second, long lasting effects of prism adaptation have been repeatedly shown in chronic (at least 3 months from injury; Frassinetti et al., 2002; Serino et al., 2006, 2007; Shiraishi et al., 2008; Vangkilde & Habekost, 2010) and sub-acute patients (at least 1 month from injury, Serino et al., 2009), whereas conflicting results have been shown when patients were treated in the acute phase (less than 1 month from injury), as in Rousseaux et al. (2006), Nys et al. (2008) and Turton et al. (2009). Acute patients generally suffer from a general decrease of attentional abilities, which may affect their ability to correctly perform a prism adaptation treatment. Furthermore, the amount and the large variability of spontaneous recovery in acute patients might overshadow specific effects of prism adaptation.

Finally, there might be important differences in the tasks used to measure prism adaptation effects. The treatment might be more effective on some visuo-spatial tasks, especially those involving a visuo-motor response, and less effective on other tasks, involving purely perceptual judgements (Striener & Danckert, 2010).

In conclusion, the present study shows that a 2-week prism adaptation treatment based on a TPA procedure, stressing a correction of visuomotor eye–hand coordinates, is more effective for ameliorating visuospatial neglect than an analogous treatment based on a CPA procedure, which primarily stresses a correction of proprioceptive coordinates. The present results may be important to reconcile some contradictory findings reported by previous studies on the effectiveness of prism adaptation treatment for neglect rehabilitation.

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